IFRS
The Department of Dry Affairs · Group 4

INTEGRATED FLOOD RESILIENCE SYSTEM

CHANNEL THE RAIN. PROTECT THE TERRAIN.

One platform that connects rooftop rainwater tanks, smart slow-release valves, storm-drain sensors and community action – piloted on the South Quay / Wrightson Road corridor in Port of Spain.

SCAN TO EXPLORE
isaacalberto.github.io/ifrs

THE PROBLEM

1960s
when much of Port of Spain's drainage was built
UN-Habitat, 2012

~30%
of drainage outfalls in central POS reported clogged
Jeppesen et al., 2015

US$11.6M
estimated annual flood damage in the target area
Inter-American Development Bank

TT$636.8M
spent on flood mitigation via PSIP, 2018–2022
Ministry of Finance

THREE DEFENCES, ONE SYSTEM

01
Rainwater collection & slow release

Roof-fed tanks catch excess rain. Sensors watch the level and a valve lets water out slowly – only when the drains downstream have room – while keeping a reserve for toilet flushing and other non-potable uses.

02
Storm-drain waste collection

Mesh basket inserts under hinged manhole covers trap litter before it blocks the drain. Drain sensors flag rising water and possible blockages so crews can act early.

03
Community empowerment

Residents report flooding and blocked drains, join cleanups, and earn reward points once staff verify their contribution.

GET STARTED IN 4 STEPS

1 Scan the code
Opens in your phone's browser. Nothing to install, no sign-up.

2 Explore the demo
Tap "Explore the live demo". You start as Aaliyah, a sample resident.

3 Switch roles
Use the name menu at the top to view as maintenance crew or an administrator.

4 Make it rain
More → Demo Controls lets you start a storm, a leak or a blocked drain and watch IFRS respond.

Samuel Blache · Asia Cooper · Keedanu Halls · Imani Zakuri

VISITOR GUIDE

MAIN FEATURES

Smart tank monitoring
See each tank's water level, capacity, free space and 48-hour history. The animated tank fills and empties with the sensor reading.

My Tanks

Safe automated release
The valve opens only when the tank is high, a set amount is kept for reuse, and the drain downstream has room. A checklist shows exactly *why* it is open or closed.

My Tanks → any tank with a valve

Drain monitoring
Live water level, how fast it is rising, basket cleaning dates and *suspected* blockages for drains along the corridor.

Drain Monitoring

Leak detection
Flags water that disappears without explanation, then guides the owner and crew through acknowledge → inspect → confirm → repair.

Leak Detection

Report a problem
Drop a pin for a flooded road, blocked drain or dumped litter, add a photo, and follow its progress from "submitted" to "resolved".

Community Action → New report

Community cleanups
Find upcoming cleanups, register, and check in on the day. Organisers verify attendance and record waste collected.

Community Action → Cleanup events

Rewards
Earn points for verified cleanups and genuine reports – each activity counts once – and redeem them for example rewards.

Rewards

Map & analytics
Every tank, drain, report and cleanup on one map, plus charts and CSV downloads. Private homes are never shown publicly.

Map & Reports

Water-rebate concept
How a future rebate programme could work: live eligibility check, estimate with its formula, application tracking and staff review.

Water Rebates

Tatil prototype station
Our test tank and drain sensor at the Tatil Building. Send a sample sensor message and watch the tank update – exactly what our Arduino board will do.

Tatil Prototype

Try these roles

Aaliyah	Resident with two home tanks, a possible leak and a pending rebate application.
Kezia	Building manager for the South Quay and Wrightson Road tanks.
Devon	Maintenance crew: verifies reports, inspects drains, works leak alerts, emergency stop.
Admin	Sets thresholds, rewards and rebate rules; manages users and devices; sees the audit log.

Quick things to try

- Open a tank and drag the simulation slider to 95% – watch the alert and valve respond.
- Demo Controls → "Storm": drains rise and valves close themselves to protect the street.
- Submit a report as Aaliyah, then switch to Devon and verify it – Aaliyah gets 25 points.

Please note

- All people, homes and readings in the app are **fictional or simulated**, except the prototype station once its sensor is connected.
- Your changes stay on your own phone. Nothing is sent to WASA, the Regional Corporation or anyone else.
- Software alone can't stop flooding – IFRS supports physical drains, tanks and maintenance.
- Real valves would be controlled by a local controller with its own safety limits, validated by qualified engineers.
- No official water-rebate scheme exists yet; the rebate section shows a proposal.
- Use **Settings → Data & privacy → Reset** to start the demo over.